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ALY 6050

Module 1 Assignment

Outcome of Baseball Series
Between the Red Sox and
Yankees



Introduction

The focus of the project is looking at probabilities and outcomes for a three-game series between the Boston Red Sox and the New York Yankees. The factors being assessed are the probabilities of the home team winning and the total bets being placed on the games in monetary values. The final question will be looking at the probabilities of each team winning if the series was stretched to five games. We will be using R to solve these calculations and the chi-squared goodness of fit tests for verification of these values.

Analysis

Part 1: If the first game is played in Boston, the second game is played in New York, and the third game (if it becomes necessary) is in Boston.

(i) Calculate the probability that the Red Sox will win the series. They have a 0.6 probability of winning when they are at home.

- R: Red Sox winning a game.
- R': Red Sox losing a game.

The Red Sox can win the series in the following ways:

- Winning the first two games.
- Losing the first game and winning the next two.
- Winning the first game, losing the second game but winning the third.

Calculating these probabilities:

- $P(R, R) = P(R) \times P(R) = 0.6 \times 0.6 = 0.36$
- $P(R', R, R) = P(R') \times P(R) \times P(R) = (1 - 0.6) \times 0.6 \times 0.6 = 0.144$
- $P(R, R', R) = P(R) \times P(R') \times P(R) = 0.6 \times (1 - 0.6) \times 0.6 = 0.144$

Adding all the probabilities for the overall probability of the Red Sox winning the series:

- Red Sox winning series = $P(R, R) + P(R', R, R) + P(R, R', R) = 0.36 + 0.144 + 0.144 = 0.648$

(ii) Construct a probability distribution for your net win (X) in the series. Calculate your expected net win (the mean of X) and the standard deviation of X:

Let's define the net win as follows:

- $X = 500$ if the Red Sox win a game.
- $X = -520$ if the Red Sox lose a game.

Now, we construct the probability distribution for X:

- $P(X = 500) = P(\text{Red Sox win}) = 0.6$
- $P(X = -520) = P(\text{Red Sox lose}) = 0.4$

```

> # Display the results
> cat("Probability of Red Sox winning the series:", prob_red_sox_win_series, "\n")
Probability of Red Sox winning the series: 0.648
> cat("Probability distribution for net win (X):\n")
Probability distribution for net win (X):
> cat("P(X = 500):", prob_X_win, "\n")
P(X = 500): 0.648
> cat("P(X = -520):", prob_X_Loss, "\n")
P(X = -520): 0.352
> cat("Expected net win (E(X)):", E_X, "\n")
Expected net win (E(X)): 140.96
> cat("Variance (Var(X)):", Var_X, "\n")
Variance (Var(X)): 237311.1
> cat("Standard deviation (SD(X)):", SD_X, "\n")
Standard deviation (SD(X)): 487.1458
>

```

(iii) Use Excel or R to create 10,000 random values for X. Let these random values be denoted by Y. Use these Y values to estimate your expected net win by using a 95% confidence interval. Does this confidence interval contain $E(X)$?

```

> # Set seed for reproducibility
> set.seed(123)
> # Generate 10,000 random values for Y
> Y <- rnorm(10000, mean = 140.96, sd = 487.1458)
> # Calculate mean and standard deviation of Y
> mean_Y <- mean(Y)
> sd_Y <- sd(Y)
> # Calculate 95% confidence interval for E(X)
> confidence_interval <- c(mean_Y - 1.96 * (sd_Y / sqrt(10000)), mean_Y + 1.96 * (sd_Y / sqrt(10000)))
> # Print results
> cat("Estimated mean of Y:", mean_Y, "\n")
Estimated mean of Y: 139.8046
> cat("95% Confidence Interval for E(X): [", confidence_interval[1], ",", confidence_interval[2], "]\n")
95% Confidence Interval for E(X): [ 130.2696 , 149.3397 ]

```

(iv) Construct a frequency distribution for Y. Next, use the Chi-squared goodness of fit test to verify how closely the distribution of Y has estimated the distribution of X.

```

#State the hypothesis of the Chi-squared test
#H0: The observed distribution of Y is the same as the expected distribution of X.
#H1: The observed distribution of Y is different from the expected distribution of X.

#Find the Critical Value
# Degrees of freedom
df <- 10

# Significance level
alpha <- 0.05

# Calculate critical value
critical_value <- qchisq(1 - alpha, df)

# Display the result
print(critical_value)
#df = 10, a = 0.05, Critical Value = 18.30704

```

```

> cat("Frequency Distribution:\n")
Frequency Distribution:
> print(frequency_table)
frequency_distribution
(-1.73e+03,-1.36e+03]      (-1.36e+03,-983]      (-983,-608]      (-608,-233]
                        8                        95                        529                        1578
      (-233,142]      (142,516]      (516,891]      (891,1.27e+03]
      2837      2752      1591      498
(1.27e+03,1.64e+03]      (1.64e+03,2.02e+03]
      98      13
> cat("\nChi-squared Test Results:\n")

Chi-squared Test Results:
> print(chi_squared_test)

Pearson's Chi-squared test

data:  observed_expected_matrix
X-squared = 1705.5, df = 10, p-value < 2.2e-16

```

(v) Use your observations of parts (ii) and (iii) above to describe whether your betting strategy is favorable to you. Write a summary of your observations and analyses in the Word document.

Considering the three scenarios of the three-game series, the Red Sox have 0.648 chance of winning the series if the first game is in Boston followed by the second game in New York and if needed the third game back in Boston. The expected net win in the series depending on if the Red Sox win or lose from the computations is \$140.96 and the standard deviation of the net win is approximately \$487.14. Factoring in all three possible outcomes of the Red Sox winning the series, they have a much better chance of winning at home so the betting strategy would depend on where the game is being played.

The confidence interval for the estimated net win was then established by creating 10,000 random samples using R. From the means of each sample the 95% confidence interval shows that the population mean is likely to lie between \$130.27 and \$149.34 and has a mean of \$139.80. Looking at the frequency distribution followed by the chi-squared goodness of fit test that was employed, the calculated x-squared value of 1705.5 is much higher than the critical value. This indicates that there is a large difference between the observed distribution and expected distribution. With a small p-value well below 0.05 as well, the null hypothesis then should be rejected.

Part 2: Repeat part 1 above but assume that the first game is played in New York, the second game is played in Boston, and the third game (if it becomes necessary) is in New York.

(i) Calculate the probability that the Yankees will win the series. They have a 0.57 probability of winning when they are at home.

- Y: Yankees winning a game.
- Y': Yankees losing a game.

The Yankees can win the series in the following ways:

- Winning the first two games.
- Losing the first game and winning the next two.

- Winning the first game, losing the second game but winning the third.

Calculating these probabilities:

- $P(Y, Y) = P(Y) \times P(Y) = 0.57 \times 0.57 = 0.3249$
- $P(Y', Y, Y) = P(Y') \times P(Y) \times P(Y) = (1 - 0.57) \times 0.57 \times 0.57 = 0.139707$
- $P(Y, Y', Y) = P(Y) \times P(Y') \times P(Y) = 0.57 \times (1 - 0.57) \times 0.57 = 0.139707$

Adding all the probabilities for the overall probability of the Yankees winning the series:

- Yankees winning series = $P(Y, Y) + P(Y', Y, Y) + P(Y, Y', Y) = 0.3249 + 0.139707 + 0.139707 = 0.604$

(ii) Construct a probability distribution for your net win (X) in the series. Calculate your expected net win (the mean of X) and the standard deviation of X:

Let's define the net win as follows:

- $X = 500$ if the Yankees win a game.
- $X = -520$ if the Yankees lose a game.

Now, we construct the probability distribution for X:

- $P(X = 500) = P(\text{Yankees win}) = 0.57$
- $P(X = -520) = P(\text{Yankees lose}) = 0.43$

```
> # Display the results
> cat("Probability of Yankees winning the series:", prob_yankees_win_series, "\n")
Probability of Yankees winning the series: 0.604
> cat("Probability distribution for net win (X):\n")
Probability distribution for net win (X):
> cat("P(X = 500):", prob_X_win, "\n")
P(X = 500): 0.604
> cat("P(X = -520):", prob_X_Loss, "\n")
P(X = -520): 0.396
> cat("Expected net win (E(X)):", E_X, "\n")
Expected net win (E(X)): 96.08
> cat("Variance (Var(X)):", var_X, "\n")
Variance (Var(X)): 248847
> cat("Standard deviation (SD(X)):", SD_X, "\n")
Standard deviation (SD(X)): 498.8457
~ |
```

(iii) Use Excel or R to create 10,000 random values for X. Let these random values be denoted by Y. Use these Y values to estimate your expected net win by using a 95% confidence interval. Does this confidence interval contain E(X)?

```

# Set seed for reproducibility
> set.seed(123)
> # Generate 10,000 random values for Y
> Y <- rnorm(10000, mean = 96.08, sd = 498.8457)
> # Calculate mean and standard deviation of Y
> mean_Y <- mean(Y)
> sd_Y <- sd(Y)
> # Calculate 95% confidence interval for E(X)
> confidence_interval <- c(mean_Y - 1.96 * (sd_Y / sqrt(10000)), mean_Y + 1.96 * (sd_Y / sqrt(10000)))
> # Print results
> cat("Estimated mean of Y:", mean_Y, "\n")
Estimated mean of Y: 94.89689
> cat("95% Confidence Interval for E(X): [", confidence_interval[1], ", ", confidence_interval[2], "]\n")
95% Confidence Interval for E(X): [ 85.13284 , 104.6609 ]

```

(iv) Construct a frequency distribution for Y. Next, use the Chi-squared goodness of fit test to verify how closely the distribution of Y has estimated the distribution of X.

```

#State the hypothesis of the Chi-squared test
#H0: The observed distribution of Y is the same as the expected distribution of X.
#H1: The observed distribution of Y is different from the expected distribution of X.

#Find the Critical Value
# Degrees of freedom
df <- 10

# Significance level
alpha <- 0.05

# Calculate critical value
critical_value <- qchisq(1 - alpha, df)

# Display the result
print(critical_value)
#df = 10, a = 0.05, Critical Value = 18.30704

> # Print results
> cat("Frequency Distribution:\n")
Frequency Distribution:
> print(frequency_table)
frequency_distribution
(-1.82e+03,-1.44e+03] (-1.44e+03,-1.05e+03] (-1.05e+03,-671] (-671,-287]
                        8                      95                      529                      1578
      (-287,96.7]      (96.7,480]      (480,864]      (864,1.25e+03]
                2837                2752                1591                498
(1.25e+03,1.63e+03] (1.63e+03,2.02e+03]
                98                  13

> cat("\nChi-squared Test Results:\n")

Chi-squared Test Results:
> print(chi_squared_test)

      Pearson's Chi-squared test

data:  observed_expected_matrix
X-squared = 1671.3, df = 10, p-value < 2.2e-16

```

(v) Use your observations of parts (ii) and (iii) above to describe whether your betting strategy is favorable to you. Write a summary of your observations and analyses in the Word document.

With the three scenarios of the three-game series, the Yankees in a home setting for the first game have 0.064 chance of winning the first game of the series followed by the second game in Boston and if needed the third game back in New York. The expected net win in the series depending on if the Yankees win or lose from the calculations is \$96.08 and the standard deviation of the net win is approximately \$498.84. While not as favorable as the Red Sox probabilities at home, it would still make sense to put money down on the Yankees in New York due to having a better chance than not of winning and therefore having better betting odds.

The confidence interval for the estimated net win was built by making 10,000 random samples with the usage of R. From the means of each sample the 95% confidence interval shows that the population mean is most likely to lie between \$85.13 and \$104.66 and contains a mean of \$94.89. The frequency distribution followed by the chi-squared goodness of fit test shows that the calculated x-squared value of 1671.3 is significantly higher than the critical value. This indicates a substantial discrepancy between the observed and expected distributions. With a small p-value well below 0.05 as well, the null hypothesis then should be rejected.

Part 3: Repeat part 1 above but now assume that the series is a best-of-five series where the first team that wins three games wins the series with games alternating between Boston and New York, with the first game being played in Boston.

(i) Calculate the probability that the Red Sox will win the series in a best-of-five series. They have a 0.6 probability of winning when they are at home.

- R: Red Sox winning a game.
- R': Red Sox losing a game.

The Red Sox can win the series in the following ways:

- Wins the first three games.
- Wins the first two games, loses the third, and wins the next one.
- Wins the first two games, loses the third and fourth, and wins the last one.
- Wins the first game, loses the second, wins the third, loses fourth, and wins the fifth.
- Wins the first game, loses the second and third, and wins the next two.
- Wins the first game, loses the second, and wins the next two.
- Loses the first two games, wins the next three.
- Loses the first game, wins the second, loses the third, and wins the next two.
- Loses the first game, wins the next three.
- Loses the first game, wins the next two, loses the fourth, wins the last.

Calculating these probabilities:

- $P(R, R, R) = P(R) \times P(R) \times P(R) = 0.6 \times 0.6 \times 0.6 = 0.216$
- $P(R', R, R, R) = P(R') \times P(R) \times P(R) \times P(R) = (1-0.6) \times 0.6 \times 0.6 \times 0.6 = 0.0864$
- $P(R, R', R, R) = P(R) \times P(R') \times P(R) \times P(R) = 0.6 \times (1-0.6) \times 0.6 \times 0.6 = 0.0864$
- $P(R, R, R', R) = P(R) \times P(R) \times P(R') \times P(R) = 0.6 \times 0.6 \times (1-0.6) \times 0.6 = 0.0864$
- $P(R, R, R', R) = P(R) \times P(R) \times P(R') \times P(R') \times P(R) = 0.6 \times 0.6 \times (1-0.6) \times (1-0.6) \times 0.6 = 0.03456$
- $P(R, R', R, R') = P(R) \times P(R') \times P(R) \times P(R') \times P(R) = 0.6 \times (1-0.6) \times 0.6 \times (1-0.6) \times 0.6 = 0.03456$
- $P(R, R', R', R, R) = P(R) \times P(R') \times P(R') \times P(R) \times P(R) = 0.6 \times (1-0.6) \times (1-0.6) \times 0.6 \times 0.6 = 0.03456$

- $P(R^-, R^-, R, R, R) = P(R^-) \times P(R^-) \times P(R) \times P(R) \times P(R) = (1-0.6) \times (1-0.6) \times 0.6 \times 0.6 \times 0.6 = 0.03456$
- $P(R^-, R, R^-, R, R) = P(R^-) \times P(R) \times P(R^-) \times P(R) \times P(R) = (1-0.6) \times 0.6 \times (1-0.6) \times 0.6 \times 0.6 = 0.03456$
- $P(R^-, R, R, R^-, R) = P(R^-) \times P(R) \times P(R) \times P(R^-) \times P(R) = (1-0.6) \times 0.6 \times 0.6 \times (1-0.6) \times 0.6 = 0.03456$

Adding all the probabilities for the overall probability of the Red Sox winning the series:

- Red Sox winning series = $P(R, R, R) + P(R^-, R, R, R) + P(R, R^-, R, R) + P(R, R, R^-, R) + P(R, R, R, R^-) + P(R, R^-, R, R^-) + P(R, R^-, R^-, R) + P(R^-, R^-, R, R) + P(R^-, R, R^-, R) + P(R^-, R, R, R^-) = 0.216 + 0.0864 + 0.0864 + 0.0864 + 0.03456 + 0.03456 + 0.03456 + 0.03456 + 0.03456 + 0.03456 = 0.682$

(ii) Construct a probability distribution for your net win (X) in the series. Calculate your expected net win (the mean of X) and the standard deviation of X:

Let's define the net win as follows:

- $X=500$ if the Red Sox win a game.
- $X=-520$ if the Red Sox lose a game.

Now, we construct the probability distribution for X:

- $P(X=500) = P(\text{Red Sox win}) = 0.682$
- $P(X=-520) = P(\text{Red Sox lose}) = 0.318$

```
> # Display the results
> cat("Probability of Red Sox winning the series in five:", prob_red_sox_win_series_five, "\n")
Probability of Red Sox winning the series in five: 0.682
> cat("Probability distribution for net win (X):\n")
Probability distribution for net win (X):
> cat("P(X = 500):", prob_X_win, "\n")
P(X = 500): 0.682
> cat("P(X = -520):", prob_X_loss, "\n")
P(X = -520): 0.318
> cat("Expected net win (E(X)):", E_X, "\n")
Expected net win (E(X)): 175.64
> cat("Variance (Var(X)):", var_X, "\n")
Variance (Var(X)): 225637.8
> cat("Standard deviation (SD(X)):", sd_X, "\n")
Standard deviation (SD(X)): 475.0135
```

(iii) Use Excel or R to create 10,000 random values for X. Let these random values be denoted by Y. Use these Y values to estimate your expected net win by using a 95% confidence interval. Does this confidence interval contain E(X)?

```
> set.seed(123)
> # Generate 10,000 random values for Y
> Y <- rnorm(10000, mean = 175.64, sd = 475.0135)
> # Calculate mean and standard deviation of Y
> mean_Y <- mean(Y)
> sd_Y <- sd(Y)
> # Calculate 95% confidence interval for E(X)
> confidence_interval <- c(mean_Y - 1.96 * (sd_Y / sqrt(10000)), mean_Y + 1.96 * (sd_Y / sqrt(10000)))
> # Print results
> cat("Estimated mean of Y:", mean_Y, "\n")
Estimated mean of Y: 174.5134
> cat("95% Confidence Interval for E(X): [", confidence_interval[1], ", ", confidence_interval[2], "]\n")
95% Confidence Interval for E(X): [ 165.2158 , 183.811 ]
```


(iv) Construct a frequency distribution for Y. Next, use the Chi-squared goodness of fit test to verify how closely the distribution of Y has estimated the distribution of X.

```
#State the hypothesis of the Chi-squared test
#H0: The observed distribution of Y is the same as the expected distribution of X.
#H1: The observed distribution of Y is different from the expected distribution of X.

#Find the Critical Value
# Degrees of freedom
df <- 10

# Significance level
alpha <- 0.05

# Calculate critical value
critical_value <- qchisq(1 - alpha, df)

# Display the result
print(critical_value)
#df = 10, a = 0.05, Critical Value = 18.30704

> # Print results
> cat("Frequency Distribution:\n")
Frequency Distribution:
> print(frequency_table)
frequency_distribution
(-1.65e+03,-1.29e+03]      (-1.29e+03,-920]      (-920,-555]      (-555,-189]
                        8                        95                        529                        1578
      (-189,176]      (176,542]      (542,907]      (907,1.27e+03]
      2837      2752      1591      498
(1.27e+03,1.64e+03]      (1.64e+03,2e+03]
      98      13
> cat("\nChi-squared Test Results:\n")

Chi-squared Test Results:
> print(chi_squared_test)

      Pearson's Chi-squared test

data:  observed_expected_matrix
X-squared = 1742.5, df = 10, p-value < 2.2e-16
```

(v) Use your observations of parts (ii) and (iii) above to describe whether your betting strategy is favorable to you. Write a summary of your observations and analyses in the Word document.

Considering the 10 scenarios of the five-game series, the Red Sox have 0.682 chance of winning the series if the first game is in Boston. Overall, there is one outcome where the team wins the first three games, three different scenarios where the series goes four games, and six different variations where the series goes five games. The expected net win in the series depending on if the Red Sox win or lose from the computations is \$175.64 and the standard deviation of the net win is approximately \$475.01. Because the expected net win is a lot higher for the five-game series, this should be heavily factored in when looking at when the Red Sox are the home team in the series.

The confidence interval for the estimated net win was created by first making 10,000 random samples with R. From the means of each sample the 95% confidence interval shows that the population mean is likely to fall in the range between \$165.22 and \$183.81 and it contains a mean of \$174.51. The frequency distribution followed by the chi-squared goodness of fit test presents the calculated x-squared value of 1742.5 which is much higher than the critical value.

This indicates a substantial difference between the observed and expected distributions. With a small p-value well below 0.05 as well, the null hypothesis then should be rejected.

Conclusion

The three possible series between the Red Sox and Yankees show that the home team that hosts game one has a higher probability of winning the series. This is consistent with the last scenario when the series is five games instead of three and the probability increases for the Red Sox who are the home team in the first game. The total net win for all three series being analyzed is also very close to the confidence intervals created, indicating that the bet totals are on par with the 10,000 random samples that were generated. This is a great example showing how bets should be predicated on the probabilities of the home team winning the first game of a three-game or five-game series.

References

ChatGPT. (2024, February 28th). Default (GPT 3.5). <https://chat.openai.com/>

Evans, J. (2012). *Statistics, Data Analysis, and Decision Modeling, 5th Edition*. Pearson Publishing.

